

Homework — 2.1.1 Thinking Abstractly

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 2.1.1 lesson.

Part A — This week's topic (~14 marks)

1. A delivery company is building software to plan the *quickest* driver routes around a city. It models the city as a **graph** (nodes joined by weighted edges).

(a) Complete the table: for each real-world detail, write **Keep** or **Discard**, deciding whether the model needs it to find the quickest route. [3] (AO2)

Detail of the city	Keep or Discard?
Locations of junctions and delivery addresses	
Typical travel time along each road	
One-way restrictions on roads	
Street names	
Colours of the buildings	
Texture of each road's surface	

(b) Explain **one** way discarding the irrelevant details helps the developers. [1] (AO2)

2. Complete the paragraph using words from the word bank. Each word may be used at most once; **three** words are not needed. [3] (AO1)

Abstraction means removing or hiding _____ detail so that a model contains only what matters for its _____. A weather app that shows the public a simple sun/cloud/rain icon is presenting a _____-level abstraction than the grid of pressure, temperature and humidity values used inside the system. A model does not have to copy reality exactly: a flight simulator keeps the _____ and instrument responses that pilots must learn, while removing the real _____ of a crash. Splitting a problem into smaller sub-problems is a different computational-thinking skill called _____.

Word bank: unnecessary · purpose · higher · lower · controls · danger · decomposition · abstraction · hardware

3. A student says: "To design this chess app I'll use abstraction — I'll split it into a move-checker, a board-display and a save-game part." The student has used the wrong term.

(a) Name the computational-thinking skill the student is actually describing, and state what that skill involves. [1] (AO1)

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(b) Give **one** genuine example of how **abstraction** could be used in the chess app. [1] (AO2)

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4. An airline trains new pilots in a **flight simulator** rather than in a real aircraft.

(a) Give **two** ways in which the simulator, as an abstraction, differs from reality. [2] (AO2)

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(b) Explain **one** reason the simulator is still effective for training pilots despite these differences. [1] (AO2)

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5. A theme park is building an app to model the queue for a ride. State **one** detail about a visitor the model should **keep** and **one** it should **discard**, justifying why the discarded detail is irrelevant to this model. [2] (AO2)

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Part B — Retrieval: keep it fresh (~6 marks)

6. (1.1.1) State the purpose of the **Program Counter (PC)** during the fetch–decode–execute cycle. [2] (AO1)

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7. (1.1.3) State **one** advantage of a **solid-state drive (SSD)** over a **magnetic hard disk drive (HDD)**. [2] (AO1)

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8. (1.1.2) State what **pipelining** is and one benefit it gives a processor. [2] (AO1)

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